- CPSC 375: High-Performance Computing

CPSC 375: High-Performance Computing

Spring 2026

- Syllabus
- Schedule
- Upload
- The Cluster Project

Assignment 1

Due 11:59 p.m., Wednesday, January 28, 2026

> **IMPORTANT!** This is an individual assignment. You may discuss broad issues of interpretation, understanding, and general approaches to a solution. However, the development of a specific solution or program code must be your own work. The assignment is expected to be entirely your own, designed and coded by you alone. If you need assistance, please consult your instructor or the TAs. Be sure to read the specific policies outlined in the **Academic Honesty in Computing** section.

Connecting Three Processes Using Pipes

Extend `whowc.c` (Slide 45, Lecture 2) to accommodate three Linux commands: Write a C program `mypipe3.c` that will take three Linux commands at the command-line as follows:

```
$ ./mypipe3 cmd1 cmd2 cmd3
```

This command should do exactly the same operation as

```
$ cmd1 | cmd2 | cmd3
```

For example, the following command will display the top 10 disk usage of your current directory:

```
$ ./mypipe3 "du" "sort -rn" "head -10"
```

which is equivalent to issuing the command:

```
$ du | sort -rn | head -10
```

Connecting *n* Processes Using Pipes

Generalize your `mypipe3.c` program to accommodate n commands, that is, write a C program, mypipen.c, so that the following command:

```
$ ./mypipen cmd1 cmd2 ... cmdn
```

does exactly the same operation as

```
$ cmd1 | cmd2 | ... | cmdn
```

Handin

When completed, submit your C source, `mypipe3.c` and `mypipen.c`, on the course website.

- **Welcome: Sean**
    - LogOut

Trinity College
HARTFORD CONNECTICUT